EPD® THE INTERNATIONAL EPD® SYSTEM

SOUTHEAST ASIA EPD THE INTERNATIONAL EPD SYSTEM

ECO PLATFORM EPD VERIFIED

Environmental Product Declaration

In accordance with **ISO 14025:2006** and

EN 15804:2012+A2:2019/AC:2021 for:

Steel Plate

This EPD of multiple products covers a range of steel plate product, based on the weighted average result of the product group across various standards applicable to steel plates.

Detailed product standards covered in this EPD can be seen in section product identification (page 5)

Programme

The International EPD® System, www.environdec.com

EPD registered through the fully aligned regional hub EPD Southeast Asia, https://www.epd-southeastasia.com/

Programme operator

EPD International AB

Regional Hub

EPD Southeast Asia

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An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com

KRAKATAU posco

PT Krakatau Posco

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General information

Programme information

Programme	**The International EPD ® System** EPD registered through the fully aligned regional hub: EPD Southeast Asia	
Address:	**EPD International AB** Box 210 60, SE-100 31 Stockholm, Sweden **EPD Southeast Asia** Kencana Tower Level M, Business Park Kebon Jeruk Jl. Raya Meruya Ilir No. 89, Jakarta Barat 11620, Indonesia	**Website** www.environdec.com www.epd-southeastasia.com **Email** info@environdec.com

Accountabilities for PCR, LCA and independent, third-party verification
Product category rules (PCR): CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): Product Category Rules (PCR) of Construction Products (PCR 2019:14 Version 1.3.4), and UN CPC 41211
PCR review was conducted by: Claudia A. Peña from PINDA LCT SpA. The review panel may be contacted via the Secretariat www.environdec.com/contact.
Life Cycle Assessment (LCA) LCA accountability: PT Life Cycle Indonesia Corresponding practitioner: Jessica Hanafi, Ph.D., Gloria FJ Kartikasari, Ruth Medyani S, Elbert Fernando, Samuel Billy B
Third-party verification Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☑ EPD verification by individual verifier
Third party verifier: Claudia A. Peña, PINDA LCT SpA (pinda.lct@gmail.com) **Approved by:** The International EPD® System Technical Committee, supported by the Secretariat
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☑ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

EPD Owner

KRAKATAU POSCO

Company information

Owner of the EPD

PT Krakatau Posco

Contact

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Description of the organisation

PT Krakatau Posco is a JVCo. (Joint Venture Company) between PT Krakatau Steel (Persero) Tbk, Indonesia and POSCO Korea. The construction commenced in 2011 and took 36 month to complete the Integrated Steel Mill with the 1st Blast Furnace Technology in Indonesia. The commercial production has started in early 2014, ready to serve Indonesia steel market and become reliable and most competitive steelmaker in regional steel market. With a current production capacity of 3 million tonnes per year, PT Krakatau Posco has the vision to be the most competitive integrated steel mill to bring pride to Indonesia. PT Krakatau Posco is expected to continue its growth. to contribute to the Indonesian economy through synergy in providing competitive steel products.

As one of the largest steel producers in Indonesia, PT Krakatau Posco not only focuses on improving production efficiency but also focuses on developing the capacity and quality of employees and environmental sustainability as evidenced by the company's commitment called PTKP 5 Global STEEL Principle.

Product Identification

PT Krakatau Posco always strives to meet consumer needs, so that PT Krakatau Posco's steel plate has met several international standards including.

American Society for Testing Material (ASTM)

- A36 Carbon Structural Steel
- A283 Low and Intermediate Tensile Strength Carbon Steel Plates
- A572 High Strength Low-Alloy Nb-V Structural Steel
- A573 Structural Carbon Steel Plate of Improved Toughness
- A709 Structural Steel for Bridge
- A285 Pressure Vessel Plates, Carbon Steel, Low and Intermediate Tensile Strength
- A515 Pressure Vessel Plates, Carbon Steel, for Intermediate and Higher Temperature Service
- A516 Pressure Vessel Plates, Carbon Steel, for Moderate and Lower Temperature Service
- A841 Steel Plates for Pressure Vessels

Japanese Industrial Standard (JIS)

- G3101-Rolled Steel for General Structure
- G3106-Rolled Steel for Welded Structure
- G3136-Rolled Steel for Building Structure

European Standard (EN)

- EN 10025-2 Non-Alloy Structural Steel
- EN 10025-3 Normalized/ Normalized rolled wieldable fine grain structural steel
- EN 10025-4 Thermomechanical rolled wieldable fine grain structural steel
- EN 10225 Weldable structural steels for fixed offshore structures

Indian Standard (IS)

- IS 2062:2011 Hot Rolled Medium and High Tensile Structural Steel

Australian – New Zealand Standard (AS/NZS)

- AS/NZS 3678:2016 Structural Steel – Hot-rolled plates, floorplates and slabs

American Petroleum Institute (API)

- API-2W Steel Pipe for Offshore Structure (API-2W 5 th Edition)
- API-5L Steel Pipe for Pipeline Transportation System (API 5L 46 th edition)

Malaysia Standard (MS)

- 10025-2 Non-Alloy Structural Steel
- 10025-3 Normalized/ Normalized rolled wieldable fine grain structural steel
- 10025-4 Thermomechanical rolled wieldable fine grain structural steel

Thailand International Standard (TIS)

- TIS 1479-Rolled Steel for General Structure
- TIS 1499-Rolled Steel for Welded Structure

Technical Standard of Viet Nam (TCVN)

- All grades that can be produced by PTKP

Shipbuilding Class

- BKI-Indonesia
- ABS-USA
- LR-UK
- DNV-Norway
- BV-France
- KR-Korea
- ClassNK-Japan
- RINA-Italy
- CR-Taiwan

Details of PT Krakatau Posco's product certification can be find on **www.krakatauposco.co.id.**

Management System Certification

PT Krakatau Posco recognises that by using an effective management system, businesses may more easily achieve and maintain high quality in both manufacturing processes and environmental performance. The following procedures and certifications can help improve the PT Krakatau Posco committee as it implements the management system.

- ISO 9001:2015 related to Quality Management System
- ISO 45001: 2018 related to Occupational Health and Safety Management System
- ISO 14001: 2015 related to Environment Management System.
- ISO/IEC 17025:2017 Laboratory Management System
- IATF 16949:2016 Automotive Quality Management System

Name and location of production site(s)

PT Krakatau Posco,
Kawasan Industri Krakatau Steel,
Jl. Afrika No.2, Kota Cilegon, Banten, 42443, Indonesia

Product information

Product name

Steel Plate

Product description

To create steel plate, PT Krakatau Posco uses computerised automatic control over the whole process. This enables us to guarantee precision and complete customisation to satisfy costumer requirements. Our steel plate's applications include shipbuilding, construction and heavy machines, offshore structures and wind farms, pressure vessels and line pipes. Key features required include chemical composition, material, dimensions, form, surface and internal quality, and additional requirements depending on the application include weldability, anti-corrosion and construction products.

UN CPC code

41211 - Flat-Rolled Products of Non-Alloy Steel, Not Further Worked than Hot-Rolled, of a Width of 600 mm or More

Geographical scope

Manufactured in Indonesia, supplied to global (the biggest customers are Indonesia, Spain, Vietnam, and Belgium).

Product identification

PT Krakatau Posco offers steel plate products that can be customized to meet the specific requirements of customers for various applications.

Product Application	Category	Standard Designation	Grade/Class	Yield Strength (MPa)	Tensile Strength (MPa)	Yield Ratio	Size	Elongation (min)
Shipbuilding steel	Mild Steel	BKI, ABS, LR, DNV, BV, KR, RINA, ClassNK, CR	A, B, D, E	235~	400~	-	6-75mmt	16%~
	High-Ten Steel	BKI, ABS, LR, DNV, BV, RINA, CR	AH32, DH32, EH32, AH36, DH36, EH36, AH40, DH40, EH40	315~	440~	-	6-75mmt	16%~
		KR, ClassNK	AH32, DH32, EH32, AH36, DH36, EH36	315~	440~	-	6-75mmt	16%~
Structural Steel		SNI	BjPS	245~	400~	-	6-75mmt	17%~
		ASTM	A36, A283 (C,D), A572 (50,60,65), A573 (70), A709 (36,50)	250~	400~	-	6-75mmt	18%~
		JIS	SS400, SS490, SS540, SM400 (A,B,C), SM490 (A,B,C), SM490 (YA,YB), SM520 (B,C), SM570, SN400 (A, B, C), SN490 (B, C)	215~	400~	-	6-75mmt	15%~
		EN & MS	S235 (JR, J0, J2), S275 (JR, J0, J2), S355 (JR, J0, J2, K2), S275 (N, NL), S355 (N, NL), S275 (M, ML), S355 (M, ML), S420 (M, ML), S460 (M, ML)	215-	360~	-	6-75mmt	22%~
		TIS	SS400, SS490, SS540, SM400 (A,B,C), SM490 (A,B,C), SM490 (YA,YB), SM520 (B,C), SM570TMC	215-	400~	-	6-75mmt	15%~
		IS	E250 (A,BR,B0,C), E275 (A,BR,B0,C), E300 (A,BR,B0,C), E350 (A,BR,B0,C)	230~	410~	-	6-75mmt	23%~
		AS	AS/NZS 250, 300, 350, 400, 450 (L0, L15, L20, L40, S0, Y20, Y40)(Z15, Z25, Z35)	250~	410~	-	6-75mmt	22%~
Pipeline Steel	Offshore		API-2W Gr.50, API-X42MO PSL 2, API-X46MO PSL 2, API-X52MO PSL 2, API-X56MO PSL 2, API-X60MO PSL 2, API-X65MO PSL 2, API-X70MO PSL 2	290~	415~	~0.93	6-35mmt	18%~
	Non-Sour		API-B PSL 1, API-X42 PSL 1, API-X46 PSL 1, API-X52 PSL 1, API-BR PSL 2, API-X42R PSL 2, API-BM PSL 2, API-X42M PSL 2, API-X46M PSL 2, API-X52M PSL 2, API-X56M PSL 2, API-X60M PSL 2, API- X65M PSL 2, API-X70M PSL 2	245~	415~	~0.93	6-35mmt	25%~
Pressure Vessel		ASTM	A285C, A515 (60, 65, 70), A516 (55, 60, 65, 70)	205~	380~	-	6-40mmt	23%~
		EN	S235 (JR, J0, J2), S275 (JR, J0, J2), S355 (JR, J0, J2, K2), S235JR, S235J2, S275JR, S275J2, S355J2, S355K2	215~	360~	-	6-75mmt	22%~

Note:

* BKI-Indonesia, ABS-USA, LR-UK, DNV-Norway, BV-France, KR-Korea, RINA-Italy, ClassNK-Japan, CR-Taiwan, SNI-Indonesia, ASTM-USA, JIS-Japan, EN-Europe, MS-Malaysia, TIS-Thailand, IS-India, AS-Australia/New Zealand

For more detailed product specifications and other grades or specifications please see **www.krakatauposco.co.id.** or contact our sales representative

LCA information

Declared unit

1 000 kg of steel plate

Reference service life

Not applicable

Time representativeness

Specific data for the manufacturing collected from 2023-01-01 to 2023-12-31. The 10-year age requirement for generic data has been met.

Database(s) and LCA software used

Generic data for upstream and downstream processes use Ecoinvent 3.10 database and modelled by using SimaPro Developer software version 9.6.0.1. No datasets older than 10 years were used.

Description of system boundaries

The system boundary was chosen based on the goal and scope of the study and in accordance with EN 15804:2012+A2:2019, i.e. “cradle-to-gate” with the end of life stage as well as benefits and loads beyond the system boundary (module A1-A4, C1-C4, D). Modules A5 and B1-B7 have not been included due to the inability to predict how the material will be used in the construction process and use stage. The processes below are included in the product system to be studied:

1 Upstream (A1-A2)

a. Production of raw materials (e.g., coking coal, iron ore, etc)

b. Production of packaging (e.g., plastic wrap, etc.)

c. Production of electricity and fuel gas (i.e. natural gas)

d. Transportation of raw/ auxiliary materials from the supplier to manufacturing plant

e. Extraction of water (i.e. tap water)

2 Core (A3)

a. Production of auxiliary materials (e.g., carbon agent, paint, etc.)

b. Coke Oven and Sinter Plant: Coke is produced from coal, and iron ore is agglomerated into sinter to improve blast furnace efficiency.

c. Blast Furnace and Steel Making: Iron ore is reduced to molten iron in the blast furnace, and impurities are removed to refine it into steel.

d. Continuous Casting: Molten steel is cast into shapes like slabs or billets for further processing.

e. Plate and Hot Strip Mills: Steel shapes are heated and rolled into plates or strips of desired thickness.

f. Marking and Packaging: Finished steel products are marked for identification and packaged for shipment.

g. Transport to waste processing unit

h. Waste processing including waste treatment process by a registered third party for hazardous waste

3 Downstream (A4, C1-C4, D)

a. A4: Transport product to customer

b. C1: Deconstruction & Demolition

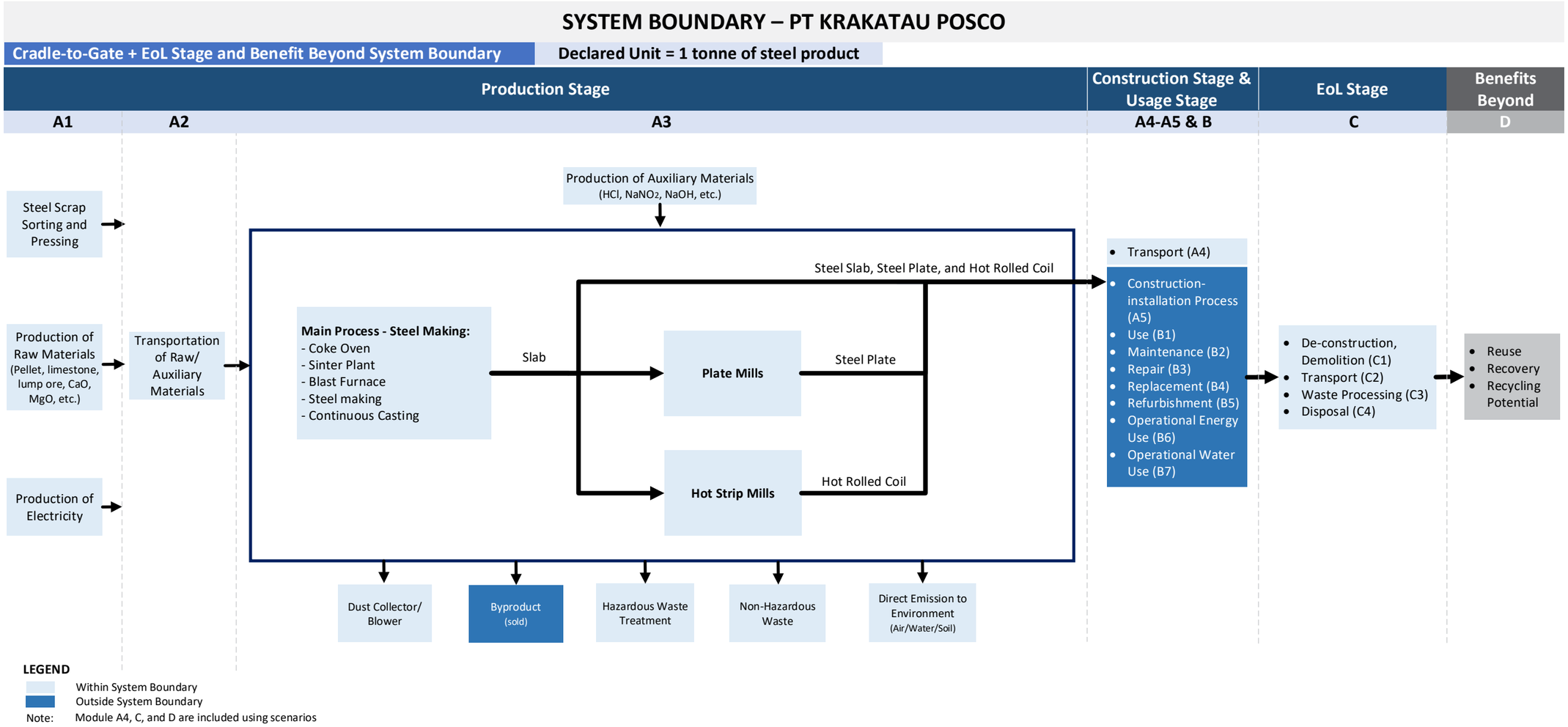
c. C2: Transport to waste processing

d. C3: Waste processing including waste treatment process by a registered third party for hazardous waste

e. C4: Disposal

f. D: Reuse/Recovery/ Recycling of the end of life of the products

System diagram



More information

Relevant websites for more information regarding the process in manufacturing:
www.krakatauposco.co.id

Key Assumptions and Limitations

a. Production process of all raw materials (i.e. coking coal, fine iron ore, anthracite, and limestone), auxiliary materials (e.g, sodium hydroxide, chloric acid, diesel, etc.), packaging (e.g., packaging seal, etc.), and natural gas use generic data from commercial databases which have been modified mainly for electricity and water according to the supply country to make it more representative.

b. The ecoinvent database provides datasets for the production of oxygen, nitrogen, and argon in liquid form, including a liquefaction process not applicable to our client's scenario. As the gases are supplied in gas phase and transported via pipeline at the client's site, the carbon dioxide values in the ecoinvent data were replaced with emission factors from the World Steel Association for gas-phase production (scope 3 emissions), ensuring accurate calculations (Janjua, 2022). The emission factors used are 0.355 $tCO_2/k.Nm^3$ for oxygen, and 0.103 $tCO_2/k.Nm^3$ for both nitrogen and argon, in line with ISO 14404:2020.

c. Electricity provided by a third party (KCE) is sourced from natural gas, combined cycle power plant. In the absence of primary data, the ecoinvent database for Indonesia's electricity dataset has been used, with the energy source composition adjusted to reflect 100% electricity production from natural gas with combined cycle power plant.

d. Excess water output in the sinter plant process, 100% is assumed to be due to loss of ignition of iron ore material that contains water and evaporates when exposed to heating treatment.

e. Economic allocation for the byproducts (tar, BTX, sulphur, heavy tar, slag, dust, and sludge) are done based on the price of the final product from the site, not based on the price of the steel product when the byproducts are produced.

f. De-construction and demolition processes are modelled using available generic data from the Ecoinvent database.

g. Waste processing and disposal of scrap steel are modelled using available generic data from commercial databases by considering what kind of waste treatment and disposal of scrap steel that occurred the most.

h. Average recovery rate for steel follows the condition in each customers' countries. Generic data from the database is used to represent the recycling facilities in each country of customers.

i. Data measured for electricity consumption are cumulative from all processes and machines.

j. In the third-party (KPE) electricity production, a small amount of steam used to generate electricity is also transferred to Krakatau Posco for heat exchange purposes. However, the inventory burden associated with electricity production, for which we have collected primary data, must inevitably be allocated 100% to the electricity, with no allocation made for the steam.

k. Data for lubricating oil, grease, refractory and battery usage for maintenance are cumulative from all machines and processes.

l. The transportation of products will serve all customer countries. Among these, 80% of the Pareto customer countries, specifically Indonesia, Spain, Vietnam, and Belgium, will cover the entire transportation system, including both sea and road transportation. The estimation for road transportation is based on the average daily distance travelled by rigid trucks, as indicated in Table 3. However, owing to the constraints posed by the limited availability of generic data, coverage for the remaining minor customer countries will be limited to sea transportation exclusively.

m. The end-of life stage (modules C1–C4) and resource recovery stage (module D) only cover 80% of Pareto rules customer countries (i.e., Indonesia, Spain, Vietnam and Belgium) due to limited generic data sources.

Cut-off rules

In accordance with EN 15804:2012+A2:2019, all available data were included in the assessment. Where input data were insufficient or data gaps existed for specific unit processes, the cut-off criteria of 1% for both renewable and non-renewable primary energy use, and 1% of the total mass input for that unit process, were applied. The total of neglected input flows did not exceed 5% of the energy usage and mass per module (e.g., modules A1–A3, C1–C4, and module D). All data relevant to the product system were considered; in cases where specific data were unavailable, appropriate generic data from databases or literature sources were used.

Data Quality

Time related coverage: specific data were collected from 2023-01-01 to 2023-12-31, and generic data are representative of the year 2021.

Geographic coverage: specific data were collected from area under study, i.e., PT Krakatau Posco, Indonesia. Generic data were collected from global average data.

Technological coverage: specific data were collected from current steel making process under study. Generic data from global average with technology aspects were similar with what described in the process under study, but merits improvement as processes were not modelled with specific data.

Data quality for both specific and generic data were sufficient to conduct life cycle assessment in accordance with the defined goal and scope

Allocation Rules

In this study, the process of continuous casting machine requires allocation as the product, i.e., slab would be either directly sold or further processed into Steel plate or Hot Rolled Coil. Mass allocation is applied mainly to Continuous Casting Machine. Economic allocation is also used in sinter plant, blast furnace, steel making, and continuous casting processes that both products undergo.

For the steel scrap used in the steel making process the end-of-waste state is when the scrap is recovered from the waste processing. The impact counted in the steel scrap used in steel making process is the transportation process from the waste processing to the GRP (post-customer). The impact of steel scrap up to the recovered material is counted in the end-of-life stage (module C).

Steel scrap can be recycled in a closed loop or open loop process, with closed loop recycling being the most common method in the steel industry. Recovered steel scrap receives a credit for avoided impacts, while scrap input incurs a debit. If the amount of recovered steel scrap for recycling is less than what the product system requires/steel scrap needed in the manufacture, then the environmental burdens associated with meeting the raw material demand are included in this closed-loop model. If, however, the amount of recovered steel scrap for recycling is larger than what the product system requires/steel scrap needed in the manufacture, then the product system receives a net credit, equivalent to the net amount of virgin material avoided.

The assessed products are distributed to Indonesia and also exported to Spain, Vietnam, and Belgium. Therefore, the recovery rate for recycling is adjusted to the rate in each country, i.e., 15% in Indonesia (Minister of National Planning and Development Indonesia, 2021), 93% in Belgium (World Auto Steel, 2024), 83.6% in Spain (Dick, 2024), and 90% in Vietnam (Amit, 2021). The remaining unrecovered steel scrap is considered as material losses that will go to other disposal scenarios to landfill. Mass allocation may be used if there are several disposal scenarios.

In this study, the closed-loop process is applied. The end-of-waste state of the steel scrap is reached when the steel scrap is processed in the waste processing (Module C3). The steel scrap is sorted and pressed into blocks and ready to be used for other specific purposes. After the point of end-of-waste, the downstream emissions related to transportation process from recycler to manufacture is attributed to the processing unit that uses the secondary material.

For the end-of-life of waste generated in the manufacturing process (e.g. contaminated rags, paint and thinner container, etc.), polluter pays principle are applied for each type of waste.

Materials sent to landfill or incineration are subject to a multi-input allocation process, where emissions are determined based on the physical/chemical properties of the material. Overhead processes associated with landfill or incineration are allocated based on the mass of the waste flows. This applies to steel scrap and waste generated in manufacturing, such as contaminated packaging and rags, also used oil.

LCA Scenarios and Additional Technical Information

Module A1-A3

- Chemical data for which there is no specific MSDS available use general data composition or generic data.
- The national grid electricity mix of Indonesia in Module A3 is based on the ecoinvent database for Indonesia electricity network which is highly reliant on coal (63%), combined cycle power plant (15%), conventional power plant (11%), hydro (6%), oil (3%), geothermal (2%), and biogas (1%). The climate impact of the electricity is 1.18 kg CO_2 eq./kWh (GWP-GHG Indicator IPCC 2021)
- Energy consumption and emissions from the transportation process of auxiliary material and waste (suppliers to manufacturing plant and transport to waste processing) are estimated using available data in the Ecoinvent database (market database).
- Air emission generated from various processes, such as Coke Oven, Sinter Plant, Blast Furnace, Steel Making, etc are based on sampling every 6 months and then extrapolated to obtain the total emissions in 1 year. The calculation is based on the flow rate of the stack, stack cross-sectional area, and running hour of the stack.
- Projected lifetime of land is assumed to be 100 years.

Module A4

- Transportation to customers in Indonesia, Spain, Vietnam, and Belgium are calculated based on the average distance travelled, weighted by the total amount of weight transported to each destination as provided by the manufacturer.

Module C

- C1: Amount of diesel used for demolition process wasmodelled using Ecoinvent database for global data, i.e.,0.626 MJ diesel/kg steel.
- C2: Transportation of steel products from warehouse to construction site, transport steel scrap from deconstruction area to waste processing unit and transport from waste processing unit to disposal site are estimated based on average rigid truck travelled kilometres a day in Indonesia (GIZ TRANSfer III, 2021), Spain (EU Statistics, 2022), Belgium (EU Statistics, 2022), and Vietnam (Lam, 2020).
- C3: Amount of diesel and electricity consumption for wasteprocessing was modelled using Ecoinvent database for global data on sorting and pressing iron scrap, i.e., 0.1MJ diesel/kg steel and 0.01 kWh/kg steel.
- C3-C4: Waste processing and disposal of scrap steel are modeled using generic data from the Ecoinvent database, assuming a 15% recycling rate in Indonesia (Minister of National Planning and Development Indonesia. 2021), 93% in Belgium (World Auto Steel, 2021), 83.60% in Spain (Dick, 2024), and 90% in Vietnam (Amit, 2021) with the remaining in each country sent to final disposal.

Module D

- The impacts assigned to the credit or burden that comes from module D are calculated by adding impact connected to secondary steel production from EAF plant (beyond system boundary) and subtracting the impacts resulting from primary steel production at BOF plant. The difference between 100% primary steel production (BOF plant) and 100% secondary steel production (EAF plant) is the result of the module D.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results)

	Product stage			Construction process stage		Use stage							End of life stage				Re-source recovery stage
	Raw material supply	Transport	Manufac-turing	Transport	Construc-tion installa-tion	Use	Mainte-nance	Repair	Replace-ment	Refurbish-ment	Opera-tional energy use	Opera-tional water use	De-con-struction demoli-tion	Transport	Waste process-ing	Disposal	Reuse-Recovery-Recycling potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	ID, RU, AU, GLO	ID, GLO	ID	ID, ES, VN, BE, GLO	-	-	-	-	-	-	-	-	ID, ES, VN, BE, GLO	ID, ES, VN, BE, GLO	ID, ES, VN, BE, GLO	ID, ES, VN, BE, GLO	ID, ES, VN, BE, GLO
Specific data used	79.98% [1]			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	<10%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-

[1]Based on GWP-GHG of stage A3 divided by GWP-GHG for stages A1-A3. Data for A3 is specific to PT Krakatau Posco Plant

Content declaration

Steel Plate manufactured by PT Krakatau Posco is made of iron ores and approximately 7% scrap-based material. PT Krakatau Posco followed the chemical range of Steel Plate as per spec, therefore, our typical composition can be seen below.

Product components	Weight, kg	Post-consumer recycled material, weight-% of product	Biogenic material, weight-% of product	Biogenic material, kg C/ product or declared unit
Scrap	70	7	0	0
Steel	930	0	0	0
Sum	**1 000**	**7**	**0**	**0**

Note: No packaging used for the products

Standard	%Carbon (C)	%Manganese (Mn)	%Silica (Si)	%Phosphorus (P)	%Sulphur (S)	%Chromium (Cr)	%Nickel (Ni)	%Copper (Cu)	%Nitrogen (N)
Range of values covering all standard and grades	0.06-0.24	0.1-1.7	0.1-0.5	0-0.025	0-0.02	0-0.35	0-0.4	0-0.035	0-0.012
Dangerous substances from the candidate list of SVHC for Authorisation									
No dangerous substances									

Results of the environmental performance indicators

- The results of this environmental impact indicator already account for infrastructure impacts in cradle and grave sourced from the generic database.
- The use of the results of modules A1-A3 without considering the results of module C is discouraged.
- The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Mandatory impact category indicators according to 15804:2012+A2:2019

Results for 1 000 kg of steel plate								
Impact Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
GWP-fossil	kg CO_2 eq.	2.44E+03	6.23E+01	6.27E+01	5.10E+01	3.17E+01	6.45E+00	-1.82E+02
GWP-biogenic	kg CO_2 eq.	6.40E-01	1.00E-02	1.00E-02	1.00E-02	1.10E-01	0.00E+00	4.88E-01
GWP-luluc	kg CO_2 eq.	8.30E-01	3.16E-02	5.45E-03	2.03E-02	4.39E-02	3.30E-03	5.00E-02
GWP-total	kg CO_2 eq.	2.45E+03	6.23E+01	6.27E+01	5.10E+01	3.18E+01	6.45E+00	-1.81E+02
ODP	kg CFC 11 eq.	9.29E-06	8.97E-07	9.60E-07	7.27E-07	4.05E-07	1.84E-07	-6.69E-06
AP	mol H^+ eq.	1.10E+01	1.58E+00	5.70E-01	2.60E-01	3.20E-01	5.00E-02	-6.40E-01
EP-freshwater	kg P eq.	2.90E-01	3.02E-04	2.21E-04	4.70E-04	2.32E-03	6.33E-05	3.92E-02
EP-marine	kg N eq.	2.83E+00	4.00E-01	2.60E-01	1.01E-01	7.00E-02	1.71E-02	-5.72E-02
EP-terrestrial	mol N eq.	3.22E+01	4.48E+00	2.87E+00	1.11E+00	8.30E-01	1.90E-01	-1.66E+00
POCP	kg NMVOC eq.	8.38E+00	1.23E+00	8.60E-01	3.63E-01	2.50E-01	7.00E-02	-1.19E+00
ADP-minerals & metals *	kg Sb eq.	2.88E-03	7.83E-05	2.24E-05	1.64E-04	1.60E-03	1.04E-05	-3.67E-03
ADP-fossil *	MJ	2.75E+04	7.85E+02	8.20E+02	7.19E+02	3.80E+02	1.56E+02	-1.46E+03
WDP *	m^3	4.18E+02	2.25E+00	1.78E+00	3.24E+00	4.74E+00	6.72E+00	-4.96E+02

* Disclaimer: The results of this environmental impact indicator may be highly uncertain in LCAs that include capital goods/infrastructure in generic datasets, in case infrastructure/capital goods contribute greatly to the total results. This is because the LCI data of infrastructure/capital goods used to quantify these indicators in currently available generic datasets sometimes lack temporal, technological and geographical representativeness. Caution should be exercised when using the results of these indicators for decision-making purposes.

- **GWP-fossil** : Global Warming Potential fossil fuels
- **GWP-biogenic** : Global Warming Potential biogenic
- **GWP-luluc** : Global Warming Potential land use and land use change
- **ODP** : Depletion potential of the stratospheric ozone layer
- **AP** : Acidification potential, Accumulated Exceedance
- **EP-freshwater** : Eutrophication potential, fraction of nutrients reaching freshwater end compartment
- **EP-marine** : Eutrophication potential, fraction of nutrients reaching marine end compartment
- **EP-terrestrial** : Eutrophication potential, Accumulated Exceedance
- **POCP** : Formation potential of tropospheric ozone
- **ADP-minerals&metals** : Abiotic depletion potential for non-fossil resources
- **ADP-fossil** : Abiotic depletion for fossil resources potential
- **WDP** : Water (user) deprivation potential, deprivation-weighted water consumption

Potential environmental impact – additional environmental information according to EN 15804:2012+A2:2019/AC:2021

Results for 1 000 kg of steel plate								
Impact Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
PM	Disease incidence	8.35E-05	2.82E-06	1.61E-05	5.07E-06	4.59E-06	1.03E-06	-6.38E-06
IRP **	kBq U235 eq.	1.87E+01	1.70E-01	1.40E-01	2.30E-01	4.10E-01	4.00E-02	7.00E-02
ETP-fw *	CTUe	1.61E+04	1.47E+02	1.16E+02	1.91E+02	2.94E+02	2.17E+01	2.48E+03
HTP-c *	CTUh	1.02E-05	2.76E-07	2.45E-07	2.73E-07	4.19E-08	2.93E-08	2.53E-06
HTP-nc *	CTUh	8.65E-06	2.68E-07	1.01E-07	5.38E-07	1.83E-06	2.82E-08	2.16E-05
SQP *	dimen-sionless	3.75E+03	1.96E+02	5.77E+01	4.28E+02	6.32E+02	3.03E+02	-8.86E+02

* Disclaimer 1: The results of this environmental impact indicator may be highly uncertain in LCAs that include capital goods/infrastructure in generic datasets, in case infrastructure/capital goods contribute greatly to the total results. This is because the LCI data of infrastructure/capital goods used to quantify these indicators in currently available generic datasets sometimes lack temporal, technological and geographical representativeness. Caution should be exercised when using the results of these indicators for decision-making purposes.

** Disclaimer 2: This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from random and from some construction materials is also not measured by this indicator.

- **PM** : Particulate Matter emissions
- **IRP** : Ionizing radiation - human healt
- **ETP-fw** : Eco-toxicity – freshwater
- **HTP-c** : Human toxicity - cancer effects
- **HTP-nc** : Human toxicity - non-cancer effects
- **SQP** : Land use related impacts / soil quality

Additional mandatory and voluntary impact category indicators

Results for 1 000 kg of steel plate								
Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
GWP-GHG	kg CO_2 eq.	2.45E+03	6.23E+01	6.27E+01	5.10E+01	3.18E+01	6.45E+00	-1.81E+02

Additional voluntary indicators e.g. the voluntary indicators from EN 15804 or the global indicators according to ISO 21930:2017

Waste production and output flows

Resource use indicators

Results for 1 000 kg of steel plate								
Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
PERE	MJ	8.00E+02	6.98E+00	5.04E+00	9.45E+00	4.62E+01	1.47E+00	-3.46E+02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	8.00E+02	6.98E+00	5.04E+00	9.45E+00	4.62E+01	1.47E+00	-3.46E+02
PENRE	MJ	2.75E+04	7.85E+02	8.20E+02	7.19E+02	3.80E+02	1.56E+02	-1.71E+03
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	2.75E+04	7.85E+02	8.20E+02	7.19E+02	3.80E+02	1.56E+02	-1.71E+03
SM	kg	1.20E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	5.69E+01	4.20E-01	4.10E-01	6.00E-01	1.09E+00	2.50E-01	1.48E+01

- **PERE** : Use of renewable primary energy excluding renewable primary energy resources used as raw materials
- **PERM** : Use of renewable primary energy resources used as raw materials
- **PERT** : Total use of renewable primary energy resources
- **PENRE** : Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials
- **PENRM** : Use of non-renewable primary energy resources used as raw materials
- **PENRT** : Total use of non-renewable primary energy re-sources
- **SM** : Use of secondary material
- **RSF** : Use of renewable secondary fuels
- **NRSF** : Use of non-renewable secondary fuels
- **FW** : Use of net fresh water

Output flow indicators

Results for 1 000 kg of steel plate								
Parameter	Unit	Total A1 -A3	A4	C1	C2	C3	C4	D
Components for re-use	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Material for recycling	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	4.00E+02	0.00E+00	0.00E+00
Materials for energy recovery	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, electricity	MJ	4.39E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, thermal	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Waste indicators

Results for 1 000 kg of steel plate								
Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
Hazardous waste disposed	kg	4.62E-01	4.44E-01	4.27E-03	5.68E-03	4.96E-03	2.01E-03	9.88E-04
Non-hazardous waste disposed	kg	1.67E+03	6.10E+02	1.31E+01	5.04E-01	3.37E+01	1.13E+01	1.00E+03
Radioactive waste disposed	kg	1.22E-02	1.15E-02	1.07E-04	9.02E-05	1.44E-04	2.79E-04	2.44E-05

Interpretation of Results

- Modules A1-A3 are the main contributors to the overall life cycle impact.
- Most significant impacts arise from processes within module A3, primarily due to the production of various supporting materials, followed by air emissions during production.
- Transportation of raw and supporting materials, especially by train, also has a substantial impact.
- Demolition and waste processing of the steel products contribute notably to the total life cycle impact.

Environmental Initiatives

PT Krakatau Posco, as one of the largest steel companies in Indonesia, has demonstrated a strong commitment to environmental conservation. Various initiatives have been undertaken to reduce the negative impact of steel production on the environment and build a more sustainable future.

Some of the environmental initiatives that have been carried out by PT Krakatau Posco include:

Steel Slag Utilization

The company is active in developing methods to recycle steel slag, which is solid waste from steel smelting process. This steel slag is then utilized as a raw material for the manufacture of environmentally friendly fertilizers that are rich in minerals. This innovation not only reduces waste, but also makes a positive contribution to the agricultural sector. In addition, by utilizing steel slag, PT Krakatau Posco is also committed to circular economy by achieving 99% waste utilization every year. Not only internally utilized, PT Krakatau Posco also collaborated with cement manufacturer to utilize steel by-products as raw materials for eco-friendly cement. This collaboration expands the market for eco-friendly products and supports sustainable development.

Clean Energy

In 2023, PT Krakatau Posco has collaborated with the biggest electricity producer in Cilegon to build solar panels on the roof of the Hot Rolling Plant. The move aims to reduce dependence on fossil energy and generate clean electricity that can be used in the production process. PT Krakatau Posco is also the only iron and steel industry in Indonesia that utilizes off-gases generated from the steel mill including Coke Oven Gas, Blast Furnace Gas, and Lintz-Donawitz Gas. This off-gas utilization effort is the most efficient way to reduce non-renewable fuel consumption in the iron and steel industry.

Environmental Care Activities

Krakatau Posco is actively involved in various social activities that focus on environmental preservation, such as beach cleanups, mangrove planting, and city park improvements. These activities involve employees and the surrounding community to raise awareness of the importance of protecting the environment. Enhancing Biodiversity: Activities such as mangrove planting help maintain coastal ecosystems and enhance biodiversity.

Through a series of comprehensive initiatives, PT Krakatau Posco is actively working to mitigate the negative environmental impacts associated with steel production. Recognizing its significant contributions to environmental sustainability, PT Krakatau Posco has been honored with prestigious awards. In 2022, the company was awarded the highest Level (Level 5) in the Green Industry Award by the Ministry of Industry, and in 2023, the company received the Green PROPER award from the Ministry of Environment and Forestry. These accolades highlight Krakatau Posco's exceptional performance in balancing economic growth with environmental responsibility. Through its commitment to environmental stewardship, Krakatau Posco not only demonstrates its dedication to sustainable practices but also contributes to improving the well-being of the surrounding communities.

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Environmental Product Declaration

KRAKATAU posco